

Power Budget

Team Number:	202						
Project Name:	CropScout						
Team Member Names:	Aaron, Jacob, Caleb, Cris, Isaiah, Asadbek, Tyler						
Version:	1						

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Buck Switching Regulator IC	AP63203WU-7	3.8 - 32V	1	2000	2000	mA
	Inductanve to Digital Converter	LDC1101DRCR	1.71 - 3.46V	1	2.7	2.7	mA
	Microcontroller	PIC18F47Q10	1.8 - 5.5V	1	400	400	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design. Add additional power rails or change the power rail voltages if needed.

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Inductanve to Digital Converter	LDC1101DRCR	1.71 - 3.46V	1	2.7	2.7	mA
	Microcontroller	PIC18F47Q10	1.8 - 5.5V	1	400	400	mA
						0	mA
						0	mA
						Subtotal	402.7 mA
						Safety Margin	25%
						Total Current Required on +3.3V Rail	503.375 mA
c4. Regulator or Source Choice	Buck Switching Regulator IC	AP63203WU-7	3.8 - 32V	1	2000	2000	mA
						Total Remaining Current Available on 3.3V Rail	1496.625 mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power sources, list each separately below and indicate which regulators will be connected to each supply. Confirm that the Total Remaining Current Available on each power source below is not negative.

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	110VAC	+24V	5000	5000	mA
Power Rails Connected to External Power Source 1	Buck Switching Regulator IC	AP63203WU-7	3.8 - 32V	3.3V	2000	2000	mA
	Total Remaining Current Available on External Power Source 1					3000	mA
External Power Source 2	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	Battery	BL5000C21704 S1PFTM104IBC 1NN	+14.4V	-14.4V	5000	5000	mA
Power Rails Connected to External Power Source 2	Buck Switching Regulator IC	AP63203WU-7	3.8 - 32V	3.3V	2000	2000	mA
	Total Remaining Current Available on External Power Source 2					3000	mA
E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.							
	Component Name	Part Number	Supply Voltage Range		Capacity (mAh)	Required By Regulators	
	Battery	BL5000C21704S1 PFTM104IBC1NN	14.4		5000	2000	
					Battery Life	2.5	hours

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit